

Neuroprotective effect of the natural iron chelator, phytic acid in a cell culture model of Parkinson's disease.

Disrupted iron metabolism and excess iron accumulation has been reported in the brains of Parkinson's disease (PD) patients. Because excessive iron can induce oxidative stress subsequently causing degradation of nigral dopaminergic neurons in PD, we determined the protective effect of a naturally occurring iron chelator, phytic acid (IP6), on 1-methyl-4-phenylpyridinium (MPP(+))-induced cell death in immortalized rat mesencephalic/dopaminergic cells. Cell death was induced with MPP(+) in normal and iron-excess conditions and cytotoxicity was measured by thiazolyl blue tetrazolium bromide (MTT assay) and trypan blue staining. Apoptotic cell death was also measured with caspase-3 activity, DNA fragmentation, and Hoechst nuclear staining. Compared to MPP(+) treatment, IP6 (30 micromol/L) increased cell viability by 19% ($P<0.05$) and decreased cell death by 22% ($P<0.05$). A threefold increase in caspase-3 activity ($P<0.001$) and a twofold increase in DNA fragmentation ($P<0.05$) with MPP(+) treatment was decreased by 55% ($P<0.01$) and 52% ($P<0.05$), respectively with IP6. Cell survival was increased by 18% ($P<0.05$) and 42% ($P<0.001$) with 30 and 100 micromol/L of IP6, respectively in iron-excess conditions. A 40% and 52% ($P<0.001$) protection was observed in caspase-3 activity with 30 and 100 micromol/L IP6, respectively in iron-excess condition. Similarly, a 45% reduction ($P<0.001$) in DNA fragmentation was found with 100 micromol/L IP6. In addition, Hoechst nuclear staining results confirmed the protective effect of IP6 against apoptosis. Similar protection was also observed with the differentiated cells. Collectively, our results demonstrate a significant neuroprotective effect of phytate in a cell culture model of PD.